



Change of Measure for Bandit Lower Bounds

How Nearby Worlds Force Every Algorithm to Gather Evidence

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A LaTeX-native research note with reproducible simulations and code.

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Chapter 1 A Lower Bound Begins by Inventing Another World

An upper bound explains what one particular algorithm can achieve. A lower bound asks for something more stubborn:

What must every algorithm pay before it can be reliably correct?

The cleanest answer begins with two possible worlds.

In the first world, arm 1 is best. In the second, arm 2 is best. The learner runs the same code in both worlds. It sees rewards one at a time, changes its sampling choices after each observation, and eventually announces an answer.

If the two worlds produce almost the same data, the learner cannot behave very differently in them. If we nevertheless demand the correct answer in both worlds, then the learner must collect enough observations to make the worlds distinguishable.

That sentence is the entire philosophy of change-of-measure lower bounds.

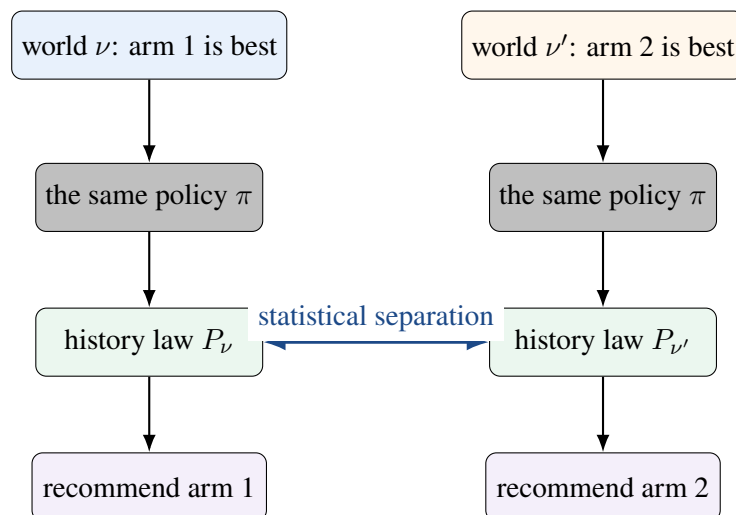


Figure 1.1: A lower bound compares the same algorithm under two environments that require different answers.

The proof strategy has three moving parts:

1. construct an alternative environment that changes the correct answer;
2. measure how much statistical evidence each observation supplies against that alternative;
3. show that reliable behavior in the two environments requires a minimum amount of total evidence.

The first step is model design. The second is information accounting. The third is hypothesis testing.

Key idea

A lower bound does not argue that an algorithm is clumsy. It argues that two worlds are too similar for *any* algorithm to separate without paying for information.

Chapter 2 One Coin, Two Possible Biases

We begin without bandits. Suppose a coin has bias either p or q , where $p \neq q$. Let

$$P = \text{Bern}(p), \quad Q = \text{Bern}(q).$$

One toss produces $X \in \{0, 1\}$.

If $X = 1$, the probability of the observation is p under P and q under Q . If $X = 0$, the probabilities are $1 - p$ and $1 - q$. The likelihood ratio is therefore

$$\frac{P(X)}{Q(X)} = \begin{cases} \frac{p}{q}, & X = 1, \\ \frac{1-p}{1-q}, & X = 0. \end{cases}$$

Its logarithm can be written without cases:

$$\ell(X) = X \log \frac{p}{q} + (1 - X) \log \frac{1-p}{1-q}. \quad (2.1)$$

A positive value favors P ; a negative value favors Q .

2.1 Evidence from several observations adds

For independent observations $X_1, \dots, X_n$,

$$\begin{aligned} \frac{P^n(X_1, \dots, X_n)}{Q^n(X_1, \dots, X_n)} &= \prod_{i=1}^n \frac{P(X_i)}{Q(X_i)}, \\ L_n := \log \frac{P^n(X_1, \dots, X_n)}{Q^n(X_1, \dots, X_n)} &= \sum_{i=1}^n \ell(X_i). \end{aligned}$$

The log-likelihood ratio is a ledger. Every observation adds one entry.

Under P , the expected entry is

$$\begin{aligned} \mathbb{E}_P[\ell(X)] &= \mathbb{E}_P \left[X \log \frac{p}{q} + (1 - X) \log \frac{1-p}{1-q} \right] \\ &= \mathbb{E}_P[X] \log \frac{p}{q} + \mathbb{E}_P[1 - X] \log \frac{1-p}{1-q} \\ &= p \log \frac{p}{q} + (1 - p) \log \frac{1-p}{1-q} \\ &= \text{kl}(p, q). \end{aligned}$$

Hence

$$\boxed{\mathbb{E}_P[L_n] = n \text{kl}(p, q).} \quad (2.2)$$

KL divergence is not merely a distance-like quantity. In this experiment it is the expected evidence supplied by one observation.

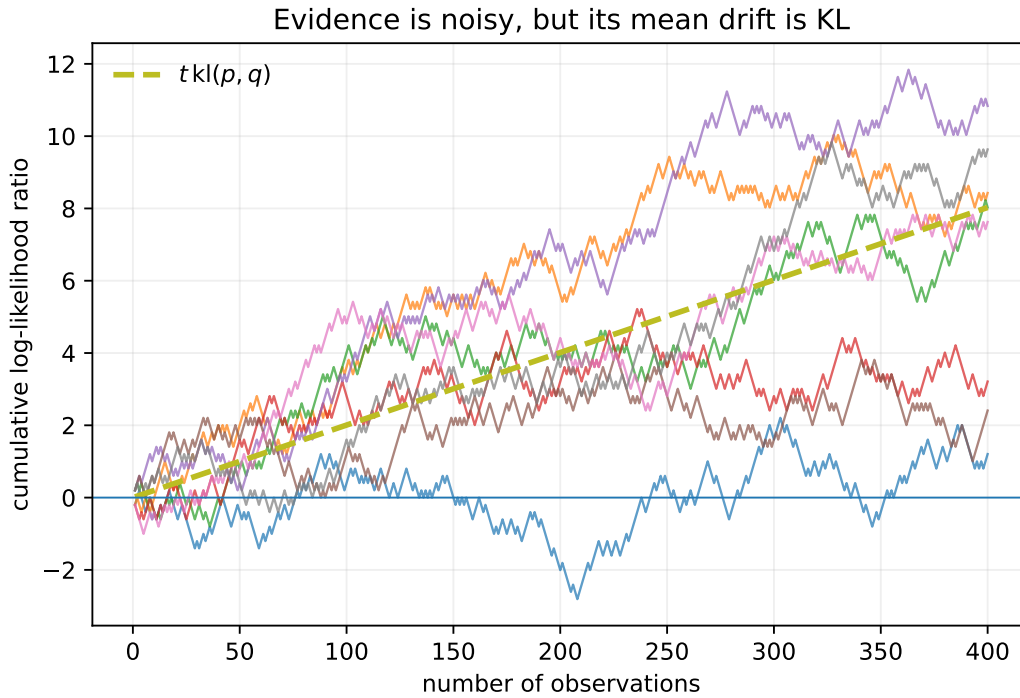


Figure 2.1: Eight evidence paths under $P = \text{Bern}(0.55)$ against $Q = \text{Bern}(0.45)$. Individual paths wander, while the expected drift is $t \text{kl}(0.55, 0.45)$.

Some paths can favor the wrong model for a long time. That is not a defect in the calculation. It is the reason a lower bound exists. Evidence arrives with noise even when its average direction is correct.

Think

If p and q move closer, then $\text{kl}(p, q)$ shrinks. The expected evidence per toss shrinks with it. A hard alternative is therefore not a wildly different world; it is a nearby world that changes the answer.

Chapter 3 Changing Measure, Without Mystery

Let P and Q be two probability laws with densities p and q on the same sample space. Define

$$L(x) = \log \frac{p(x)}{q(x)}.$$

Then

$$q(x) = p(x)e^{-L(x)}.$$

For any event E ,

$$\begin{aligned} Q(E) &= \int_E q(x) \, dx \\ &= \int_E p(x)e^{-L(x)} \, dx \\ &= \mathbb{E}_P [\mathbf{1}_E e^{-L}]. \end{aligned}$$

Thus

$$\boxed{Q(E) = \mathbb{E}_P [\mathbf{1}_E e^{-L}]} \quad (3.1)$$

This is the change-of-measure identity. It says that a probability in the Q -world can be computed by averaging in the P -world, provided each outcome is reweighted by how much more or less plausible it is under Q .

There is no algorithm in this identity. It is a statement about two probability laws. Bandit lower bounds become possible when we apply it to the law of an entire adaptive history.

3.1 Compressing the whole experiment to one event

Suppose the final decision is summarized by an event E . Let

$$\alpha = P(E), \quad \beta = Q(E).$$

The entire observation may be complicated, but the event records only one bit: did E occur?

Information cannot increase when data are compressed. In this binary case,

$$\boxed{\text{KL}(P\|Q) \geq \text{kl}(\alpha, \beta)} \quad (3.2)$$

We now derive it directly.

On the event E , write P_E and Q_E for the conditional laws. Then

$$P(dx) = \alpha P_E(dx), \quad Q(dx) = \beta Q_E(dx), \quad x \in E.$$

Therefore, on E ,

$$\frac{dP}{dQ} = \frac{\alpha}{\beta} \frac{dP_E}{dQ_E}.$$

The contribution of E to KL is

$$\begin{aligned} \int_E \log \frac{dP}{dQ} \, dP &= \alpha \int_E \log \left(\frac{\alpha}{\beta} \frac{dP_E}{dQ_E} \right) \, dP_E \\ &= \alpha \log \frac{\alpha}{\beta} + \alpha \text{KL}(P_E\|Q_E). \end{aligned}$$

The same calculation on E^c gives

$$(1 - \alpha) \log \frac{1 - \alpha}{1 - \beta} + (1 - \alpha) \text{KL}(P_{E^c}\|Q_{E^c}).$$

Adding both pieces,

$$\begin{aligned}
 \text{KL}(P\|Q) &= \alpha \log \frac{\alpha}{\beta} + (1 - \alpha) \log \frac{1 - \alpha}{1 - \beta} \\
 &\quad + \alpha \text{KL}(P_E\|Q_E) + (1 - \alpha) \text{KL}(P_{E^c}\|Q_{E^c}) \\
 &\geq \alpha \log \frac{\alpha}{\beta} + (1 - \alpha) \log \frac{1 - \alpha}{1 - \beta} \\
 &= \text{kl}(\alpha, \beta).
 \end{aligned}$$

The inequality uses only the nonnegativity of KL divergence.

Key idea

A lower-bound proof does not need to understand every feature of the final history. It can compress the history to the one event on which the two worlds must disagree.

3.2 A useful testing inequality

Another common form is the Bretagnolle–Huber inequality. For every event E ,

$$P(E) + Q(E^c) \geq \frac{1}{2} \exp\{-\text{KL}(P\|Q)\}. \quad (3.3)$$

Interpret E as the decision “choose Q .” Then $P(E)$ is an error under P , while $Q(E^c)$ is an error under Q . Equation (3.3) says that the two errors cannot both be tiny unless the two laws have accumulated substantial KL divergence.

The complete proof is given in Appendix 12. For now, the important shape is

$$\text{testing error} \gtrsim e^{-\text{information}}.$$

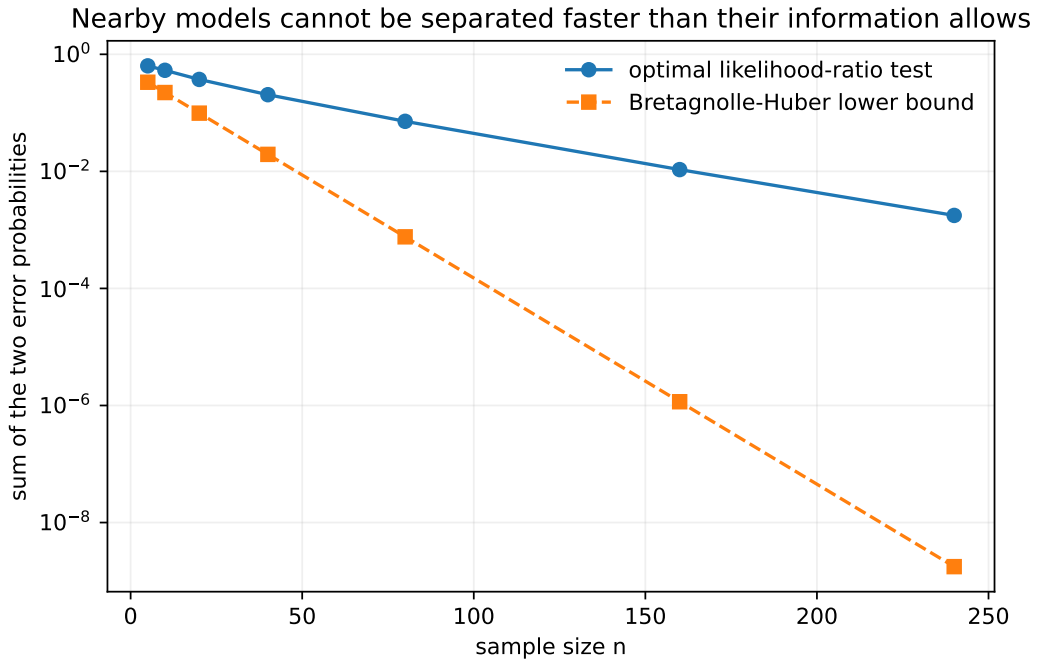


Figure 3.1: The exact error of the optimal likelihood-ratio test and the Bretagnolle–Huber lower bound for $\text{Bern}(0.4)$ versus $\text{Bern}(0.6)$. The bound is not exact, but it captures the unavoidable exponential scale.

The bound is deliberately simple and therefore not always numerically tight. Its value is portability: the same inequality can be attached to an adaptive history, a stopping rule, or a recommendation event without redesigning the test from scratch.

Chapter 4 The Law of an Adaptive Bandit History

Return to a K -armed bandit. Let

$$\nu = (\nu_1, \dots, \nu_K)$$

be one environment and

$$\nu' = (\nu'_1, \dots, \nu'_K)$$

be another. The policy is the same in both environments.

At round t , the policy chooses an arm A_t from the past history

$$H_{t-1} = (A_1, X_1, \dots, A_{t-1}, X_{t-1}).$$

Write

$$\pi_t(a \mid h_{t-1})$$

for the probability that the policy chooses arm a after history h_{t-1} . If arm a is chosen, its reward density is p_a under ν and p'_a under ν' .

For a realized history

$$h_T = (a_1, x_1, \dots, a_T, x_T),$$

the density under ν factors as

$$P_\nu(h_T) = \prod_{t=1}^T \pi_t(a_t \mid h_{t-1}) p_{a_t}(x_t). \quad (4.1)$$

Under ν' ,

$$P_{\nu'}(h_T) = \prod_{t=1}^T \pi_t(a_t \mid h_{t-1}) p'_{a_t}(x_t). \quad (4.2)$$

4.1 The policy terms cancel

Divide (4.1) by (4.2):

$$\begin{aligned} \frac{P_\nu(h_T)}{P_{\nu'}(h_T)} &= \frac{\prod_{t=1}^T \pi_t(a_t \mid h_{t-1}) p_{a_t}(x_t)}{\prod_{t=1}^T \pi_t(a_t \mid h_{t-1}) p'_{a_t}(x_t)} \\ &= \prod_{t=1}^T \frac{p_{a_t}(x_t)}{p'_{a_t}(x_t)}. \end{aligned}$$

Therefore the history log-likelihood ratio is

$$L_T = \sum_{t=1}^T \log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)}. \quad (4.3)$$

This cancellation is the central algebraic fact. The policy may be adaptive, randomized, and highly non-linear. None of those policy probabilities appears in the final likelihood ratio, because the same decision rule is used in both worlds.

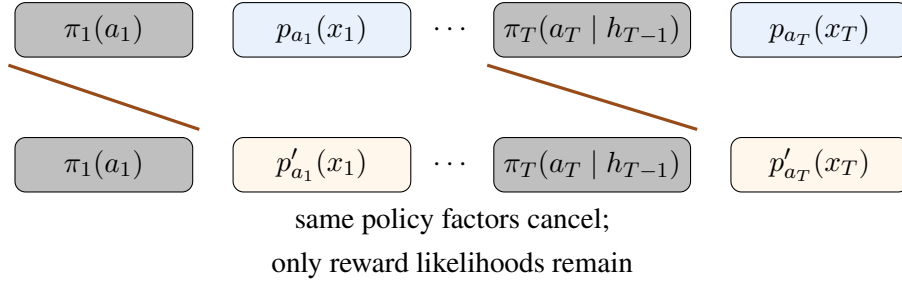


Figure 4.1: The learner’s adaptivity changes which reward factors appear, but not the likelihood-ratio algebra.

4.2 Expected evidence equals pulls times armwise KL

Take expectations under ν . From (4.3),

$$\mathbb{E}_\nu[L_T] = \sum_{t=1}^T \mathbb{E}_\nu \left[\log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)} \right].$$

Condition on the selected arm. If $A_t = a$, then $X_t \sim \nu_a$ under ν , and therefore

$$\begin{aligned} \mathbb{E}_\nu \left[\log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)} \middle| A_t = a, H_{t-1} \right] &= \int p_a(x) \log \frac{p_a(x)}{p'_a(x)} dx \\ &= \text{KL}(\nu_a \| \nu'_a). \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}_\nu[L_T] &= \sum_{t=1}^T \mathbb{E}_\nu [\text{KL}(\nu_{A_t} \| \nu'_{A_t})] \\ &= \sum_{t=1}^T \sum_{a=1}^K \mathbb{P}_\nu(A_t = a) \text{KL}(\nu_a \| \nu'_a) \\ &= \sum_{a=1}^K \mathbb{E}_\nu \left[\sum_{t=1}^T \mathbf{1}\{A_t = a\} \right] \text{KL}(\nu_a \| \nu'_a). \end{aligned}$$

Define the number of pulls

$$N_a(T) = \sum_{t=1}^T \mathbf{1}\{A_t = a\}.$$

Then

$$\boxed{\text{KL}(P_\nu^T \| P_{\nu'}^T) = \mathbb{E}_\nu[L_T] = \sum_{a=1}^K \mathbb{E}_\nu[N_a(T)] \text{KL}(\nu_a \| \nu'_a).} \quad (4.4)$$

This is often called the divergence decomposition. It turns a complicated adaptive experiment into a simple information ledger:

$$\text{expected pulls of arm } a \quad \times \quad \text{information supplied by one pull of arm } a.$$

Key idea

Adaptivity changes the random pull counts. It does not create information from nowhere. Every unit of KL must still be paid for by observations from arms on which the two environments differ.

4.3 A numerical audit of the identity

The supplied code runs UCB under

$$\nu = (\text{Bern}(0.62), \text{Bern}(0.50))$$

and compares it with the alternative

$$\nu' = (\text{Bern}(0.62), \text{Bern}(0.68)).$$

Only arm 2 changes. For each simulated history we compute the realized log-likelihood ratio and compare its average with

$$\mathbb{E}_\nu[N_2(T)] \text{kl}(0.50, 0.68).$$

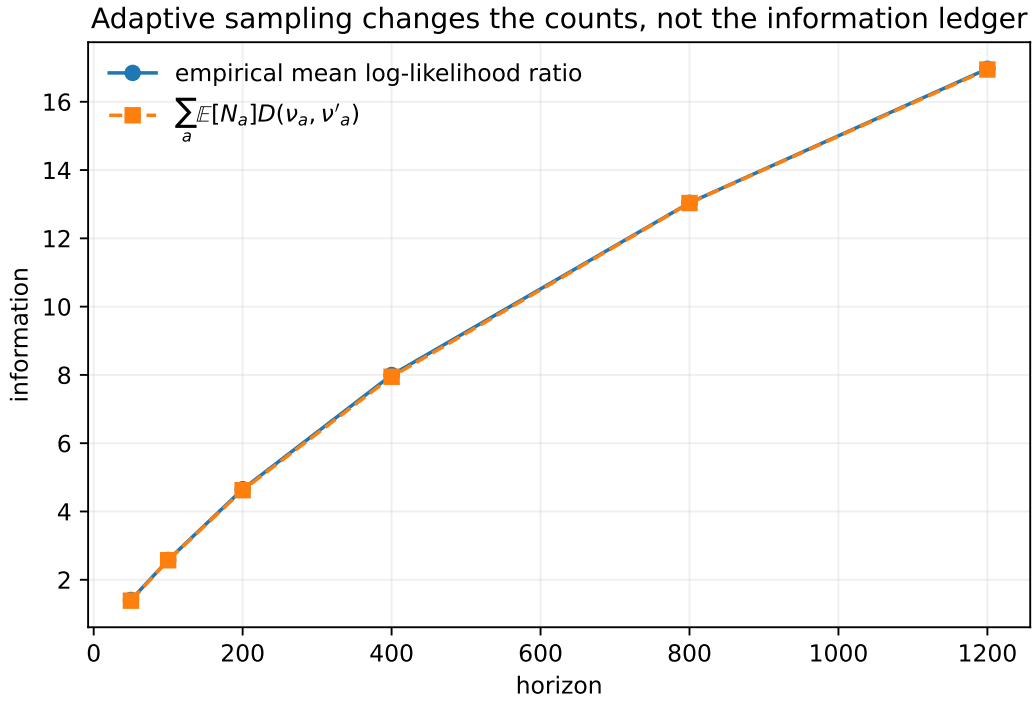


Figure 4.2: The empirical mean log-likelihood ratio matches the pull-count information ledger, even though the sampling rule is adaptive.

Table 4.1: Adaptive information accounting under UCB, based on 12,000 replications.

Horizon	$\mathbb{E}[N_2(T)]$	empirical $\mathbb{E}[L_T]$	$\mathbb{E}[N_2(T)] \text{kl}(0.50, 0.68)$
100	37.05	2.581	2.572
400	114.50	7.997	7.946
800	187.81	13.040	13.034
1200	244.16	16.966	16.945

The code performing the essential update is short:

```
1 means = sums / counts
2 bonus = np.sqrt(2.0 * np.log(t + 1.0) / counts)
```

```
3 actions = np.argmax(means + bonus, axis=1)
4
5 arm_means = true_means[actions]
6 alt_means = alternative_means[actions]
7 rewards = rng.binomial(1, arm_means).astype(float)
8
9 counts[indices, actions] += 1
10 sums[indices, actions] += rewards
11
12 total_llr += (
13     rewards * np.log(arm_means / alt_means)
14     + (1.0 - rewards)
15     * np.log((1.0 - arm_means) / (1.0 - alt_means))
16 )
```

Chapter 5 Stopping When the Evidence Is Enough

Many bandit algorithms do not stop at a fixed horizon. They stop when the data appear decisive. Let τ be such a stopping time, and let $\mathcal{F}_\tau$ contain everything known when the algorithm stops.

The stopped log-likelihood ratio is

$$L_\tau = \sum_{t=1}^{\tau} \log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)}.$$

Under the usual integrability conditions, the same information ledger holds:

$$\mathbb{E}_\nu[L_\tau] = \sum_{a=1}^K \mathbb{E}_\nu[N_a(\tau)] \text{KL}(\nu_a \| \nu'_a). \quad (5.1)$$

Now take any event $E \in \mathcal{F}_\tau$. It could be the event that the algorithm recommends arm 1, eliminates arm 4, or stops before a given time. Applying binary data processing to the stopped history gives

$$\mathbb{E}_\nu[L_\tau] \geq \text{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E)).$$

Combining with (5.1) yields the fundamental bandit change-of-measure inequality.

Bandit transportation inequality

For two bandit environments ν and ν' , an almost surely finite stopping time τ , and any event $E \in \mathcal{F}_\tau$,

$$\sum_{a=1}^K \mathbb{E}_\nu[N_a(\tau)] \text{KL}(\nu_a \| \nu'_a) \geq \text{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E)). \quad (5.2)$$

The word “transportation” is helpful. The left side measures how much information the algorithm transports from observations into its history. The right side measures how far apart the algorithm’s behavior must be in the two worlds.

5.1 The proof in three transparent steps

Step 1: the stopped change-of-measure identity. For $E \in \mathcal{F}_\tau$,

$$\mathbb{P}_{\nu'}(E) = \mathbb{E}_\nu [\mathbf{1}_E e^{-L_\tau}].$$

Step 2: compress the stopped history to E . The laws of the whole stopped history satisfy

$$\text{KL}(P_\nu^{H_\tau} \| P_{\nu'}^{H_\tau}) \geq \text{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E)).$$

Since the log density ratio of the stopped histories is L_τ ,

$$\text{KL}(P_\nu^{H_\tau} \| P_{\nu'}^{H_\tau}) = \mathbb{E}_\nu[L_\tau].$$

Step 3: expand the expected evidence arm by arm. Write the observations from arm a in their order of appearance as

$$Y_{a,1}, Y_{a,2}, \dots$$

Then

$$L_\tau = \sum_{a=1}^K \sum_{s=1}^{N_a(\tau)} \log \frac{p_a(Y_{a,s})}{p'_a(Y_{a,s})}.$$

The expected increment from arm a is

$$\mathbb{E}_\nu \left[\log \frac{p_a(Y_{a,s})}{p'_a(Y_{a,s})} \right] = \text{KL}(\nu_a \| \nu'_a).$$

Wald's identity therefore gives

$$\mathbb{E}_\nu[L_\tau] = \sum_{a=1}^K \mathbb{E}_\nu[N_a(\tau)] \text{KL}(\nu_a \| \nu'_a).$$

Putting the three steps together proves (5.2).

Proof pattern

A change-of-measure lower bound usually has the same skeleton:

sample allocation $\longrightarrow$ KL information $\longrightarrow$ separation of a decision event.

The creativity lies mainly in choosing the alternative environment and the event.

Chapter 6 The Simplest Sample-Complexity Lower Bound

Return once more to the two-coin problem. Suppose a decision rule observes n tosses and must identify whether the coin is $P = \text{Bern}(p)$ or $Q = \text{Bern}(q)$. Let

$$E = \{\text{the rule declares } P\}.$$

Assume the rule has error at most δ in both worlds:

$$P^n(E) \geq 1 - \delta, \quad Q^n(E) \leq \delta.$$

By (3.2),

$$\begin{aligned} \text{KL}(P^n \| Q^n) &\geq \text{kl}(P^n(E), Q^n(E)) \\ &\geq \text{kl}(1 - \delta, \delta). \end{aligned}$$

Independence gives

$$\text{KL}(P^n \| Q^n) = n \text{KL}(P \| Q) = n \text{kl}(p, q).$$

Therefore

$$\boxed{n \geq \frac{\text{kl}(1 - \delta, \delta)}{\text{kl}(p, q)}}. \tag{6.1}$$

Every part has an interpretation:

$$\underbrace{n}_{\text{number of observations}} \times \underbrace{\text{kl}(p, q)}_{\text{evidence per observation}} \geq \underbrace{\text{kl}(1 - \delta, \delta)}_{\text{evidence demanded by reliability}}.$$

For $p = 0.55$, $q = 0.45$, and $\delta = 0.05$,

$$\text{kl}(0.55, 0.45) \approx 0.02007, \quad \text{kl}(0.95, 0.05) \approx 2.650.$$

Thus

$$n \geq \frac{2.650}{0.02007} \approx 132.1.$$

A rule that promises five-percent error in both directions needs at least about 133 tosses according to this information bound.

As $\delta \downarrow 0$,

$$\text{kl}(1 - \delta, \delta) = (1 - 2\delta) \log \frac{1 - \delta}{\delta} \sim \log \frac{1}{\delta}.$$

So the familiar logarithm in fixed-confidence sample complexity is not an artifact of one algorithm. It is the amount of evidence required to drive an error probability down to δ .

Chapter 7 Best-Arm Identification: Make the Answer Change

Consider a bandit environment ν with a unique best arm

$$a^*(\nu) = \arg \max_a \mu_a.$$

A fixed-confidence algorithm stops at τ and recommends $\hat{a}$. It is δ -correct if

$$\mathbb{P}_\nu(\hat{a} = a^*(\nu)) \geq 1 - \delta$$

for every environment in the model class.

Fix the true environment ν . Now choose an alternative λ whose best arm is different:

$$a^*(\lambda) \neq a^*(\nu).$$

Define

$$E = \{\hat{a} = a^*(\nu)\}.$$

Correctness in the true world gives

$$\mathbb{P}_\nu(E) \geq 1 - \delta.$$

Correctness in the alternative world gives

$$\mathbb{P}_\lambda(E) \leq \delta,$$

because the arm recommended on E is wrong under λ .

Insert this event into the transportation inequality:

$$\sum_{a=1}^K \mathbb{E}_\nu[N_a(\tau)] \text{KL}(\nu_a \parallel \lambda_a) \geq \text{kl}(1 - \delta, \delta). \quad (7.1)$$

This inequality holds for *every* alternative that changes the best arm.

7.1 A first, deliberately simple alternative

Suppose there are two Bernoulli arms with

$$\mu_1 > \mu_2.$$

Keep arm 1 unchanged and raise arm 2 to a mean $\lambda_2 > \mu_1$. Then only arm 2 differs, and (7.1) becomes

$$\mathbb{E}_\nu[N_2(\tau)] \text{kl}(\mu_2, \lambda_2) \geq \text{kl}(1 - \delta, \delta).$$

Hence

$$\mathbb{E}_\nu[N_2(\tau)] \geq \frac{\text{kl}(1 - \delta, \delta)}{\text{kl}(\mu_2, \lambda_2)}. \quad (7.2)$$

Let λ_2 approach μ_1 from above. The alternative becomes as close as possible while still changing the best arm, yielding

$$\mathbb{E}_\nu[N_2(\tau)] \gtrsim \frac{\text{kl}(1 - \delta, \delta)}{\text{kl}(\mu_2, \mu_1)}.$$

This already explains why a suboptimal arm cannot simply be ignored. If the learner rarely samples arm 2, it cannot rule out the nearby world in which arm 2 is actually best.

Think

Why not raise arm 2 to 0.99? Because that alternative is easy to distinguish. Its KL divergence is large, so the resulting lower bound is weak. Lower bounds become strong by finding the *closest answer-changing alternative*.

7.2 The allocation game

Let

$$\text{Alt}(\nu) = \{\lambda : a^*(\lambda) \neq a^*(\nu)\}$$

be the set of answer-changing alternatives. Define the expected allocation proportions

$$w_a = \frac{\mathbb{E}_\nu[N_a(\tau)]}{\mathbb{E}_\nu[\tau]}, \quad \sum_{a=1}^K w_a = 1.$$

Then (7.1) can be written

$$\mathbb{E}_\nu[\tau] \sum_{a=1}^K w_a \text{KL}(\nu_a \| \lambda_a) \geq \text{kl}(1 - \delta, \delta).$$

Because this must hold for every $\lambda \in \text{Alt}(\nu)$,

$$\mathbb{E}_\nu[\tau] \inf_{\lambda \in \text{Alt}(\nu)} \sum_{a=1}^K w_a \text{KL}(\nu_a \| \lambda_a) \geq \text{kl}(1 - \delta, \delta).$$

Thus

$$\mathbb{E}_\nu[\tau] \geq \frac{\text{kl}(1 - \delta, \delta)}{\inf_{\lambda \in \text{Alt}(\nu)} \sum_{a=1}^K w_a \text{KL}(\nu_a \| \lambda_a)}. \quad (7.3)$$

The algorithm chooses w . The lower-bound argument then chooses the hardest alternative against that allocation. This leads to the characteristic information value

$$\frac{1}{T^*(\nu)} = \sup_{w \in \Delta_K} \inf_{\lambda \in \text{Alt}(\nu)} \sum_{a=1}^K w_a \text{KL}(\nu_a \| \lambda_a), \quad (7.4)$$

where

$$\Delta_K = \left\{ w \in [0, 1]^K : \sum_{a=1}^K w_a = 1 \right\}.$$

Consequently,

$$\mathbb{E}_\nu[\tau] \geq T^*(\nu) \text{kl}(1 - \delta, \delta). \quad (7.5)$$

Research lens

Equation (7.4) is not only a lower bound. It is an algorithm-design target. Track-and-Stop and later best-arm identification methods attempt to learn and track the allocation that maximizes the worst-case information rate. The same information-allocation viewpoint also underlies modern batched BAI and regret-aware BAI, including research directions developed by Tianyuan Jin and collaborators.

Chapter 8 Two Gaussian Arms: Solve the Lower-Bound Game Completely

Consider two Gaussian arms with known common variance σ^2 :

$$\nu_1 = \mathcal{N}(\mu_1, \sigma^2), \quad \nu_2 = \mathcal{N}(\mu_2, \sigma^2), \quad \mu_1 > \mu_2.$$

Let

$$\Delta = \mu_1 - \mu_2 > 0.$$

Assign a fraction w of samples to arm 1 and $1 - w$ to arm 2.

For an alternative mean vector (λ_1, λ_2) , Gaussian KL gives

$$\text{KL}(\mathcal{N}(\mu_a, \sigma^2) \parallel \mathcal{N}(\lambda_a, \sigma^2)) = \frac{(\mu_a - \lambda_a)^2}{2\sigma^2}.$$

The information rate is therefore

$$I_w(\lambda_1, \lambda_2) = \frac{w(\mu_1 - \lambda_1)^2 + (1 - w)(\mu_2 - \lambda_2)^2}{2\sigma^2}. \quad (8.1)$$

The alternative must make arm 2 at least as good as arm 1:

$$\lambda_1 \leq \lambda_2.$$

The closest point in this half-space lies on the boundary

$$\lambda_1 = \lambda_2 = m.$$

Thus we minimize

$$I_w(m) = \frac{w(\mu_1 - m)^2 + (1 - w)(\mu_2 - m)^2}{2\sigma^2}.$$

8.1 Step 1: find the hardest common mean

Differentiate with respect to m :

$$\frac{d}{dm} [w(\mu_1 - m)^2 + (1 - w)(\mu_2 - m)^2] = 2w(m - \mu_1) + 2(1 - w)(m - \mu_2).$$

Set this equal to zero:

$$\begin{aligned} 0 &= w(m - \mu_1) + (1 - w)(m - \mu_2) \\ &= m - w\mu_1 - (1 - w)\mu_2. \end{aligned}$$

Therefore

$$\boxed{m_w = w\mu_1 + (1 - w)\mu_2}. \quad (8.2)$$

The hard alternative moves toward the arm receiving fewer samples. If arm 1 receives most of the budget, then its mean is well measured and the alternative moves close to μ_1 , leaving the poorly measured arm 2 to do most of the work.

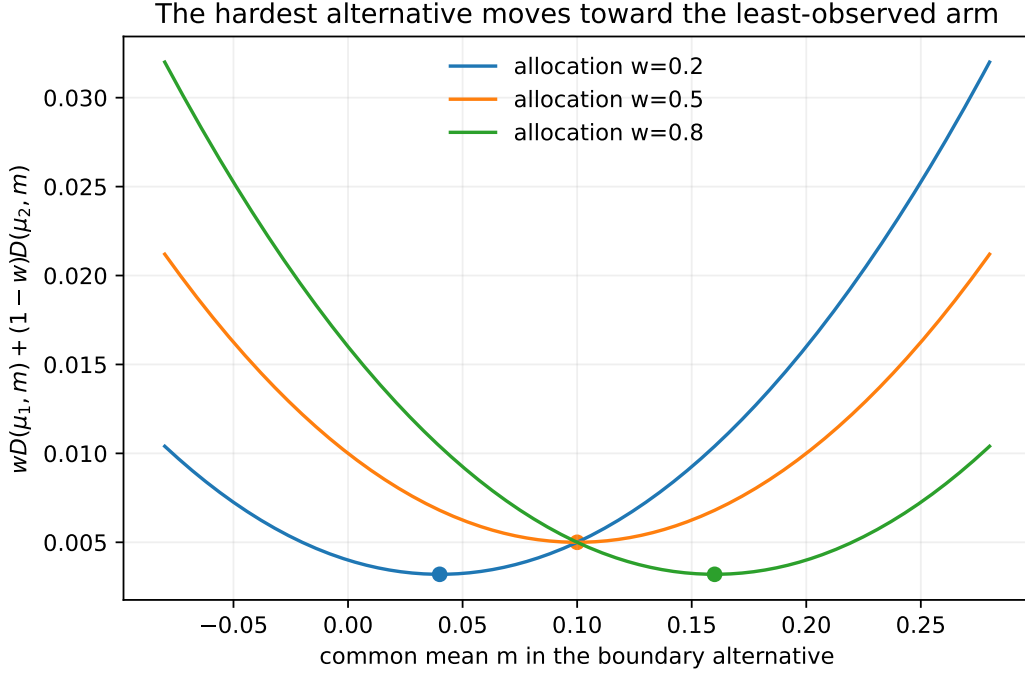


Figure 8.1: For each allocation w , the hardest boundary alternative is the minimum of the corresponding quadratic information curve.

8.2 Step 2: substitute the minimizer

From (8.2),

$$\begin{aligned}\mu_1 - m_w &= \mu_1 - w\mu_1 - (1-w)\mu_2 \\ &= (1-w)(\mu_1 - \mu_2) \\ &= (1-w)\Delta,\end{aligned}$$

and

$$\begin{aligned}\mu_2 - m_w &= \mu_2 - w\mu_1 - (1-w)\mu_2 \\ &= -w(\mu_1 - \mu_2) \\ &= -w\Delta.\end{aligned}$$

Therefore

$$\begin{aligned}\inf_{\lambda_1 \leq \lambda_2} I_w(\lambda_1, \lambda_2) &= \frac{w(1-w)^2\Delta^2 + (1-w)w^2\Delta^2}{2\sigma^2} \\ &= \frac{w(1-w)((1-w) + w)\Delta^2}{2\sigma^2} \\ &= \frac{w(1-w)\Delta^2}{2\sigma^2}.\end{aligned}$$

8.3 Step 3: choose the best allocation

Since

$$w(1-w) = \frac{1}{4} - \left(w - \frac{1}{2}\right)^2,$$

the maximum occurs at

$$w^* = \frac{1}{2}.$$

The optimal information rate is

$$\frac{1}{T^*(\nu)} = \frac{\Delta^2}{8\sigma^2}. \quad (8.3)$$

Hence

$$\mathbb{E}_\nu[\tau] \geq \frac{8\sigma^2}{\Delta^2} \text{kl}(1 - \delta, \delta). \quad (8.4)$$

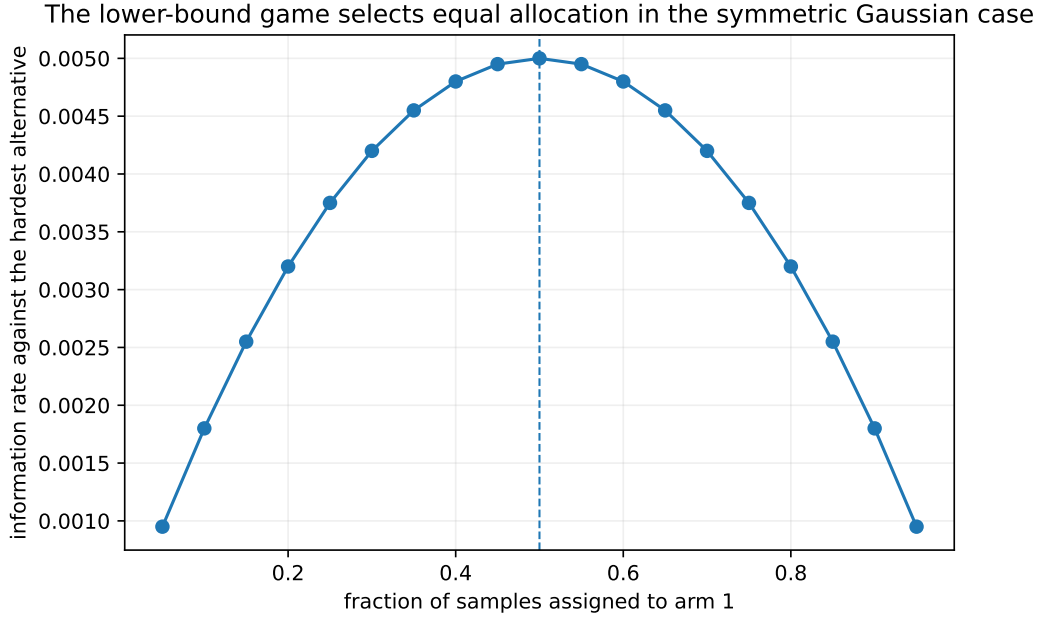


Figure 8.2: The worst-case information rate is maximized by equal allocation in the symmetric two-Gaussian problem.

This conclusion is easy to say after the calculation: both sample means enter the comparison, so making one estimate precise while leaving the other noisy is wasteful. The lower-bound game turns that intuition into an exact constant.

8.4 The same allocation appears in the actual error

With a fixed total budget n , let

$$n_1 = wn, \quad n_2 = (1 - w)n.$$

The sample-mean difference satisfies

$$\hat{\mu}_1 - \hat{\mu}_2 \sim \mathcal{N}\left(\Delta, \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right).$$

If the learner recommends the larger sample mean, then

$$\begin{aligned} \mathbb{P}(\text{error}) &= \mathbb{P}(\hat{\mu}_1 - \hat{\mu}_2 \leq 0) \\ &= \Phi\left(-\frac{\Delta}{\sigma\sqrt{1/n_1 + 1/n_2}}\right). \end{aligned}$$

The denominator is minimized at $n_1 = n_2$. Thus the finite-sample testing calculation and the information lower bound select the same allocation.

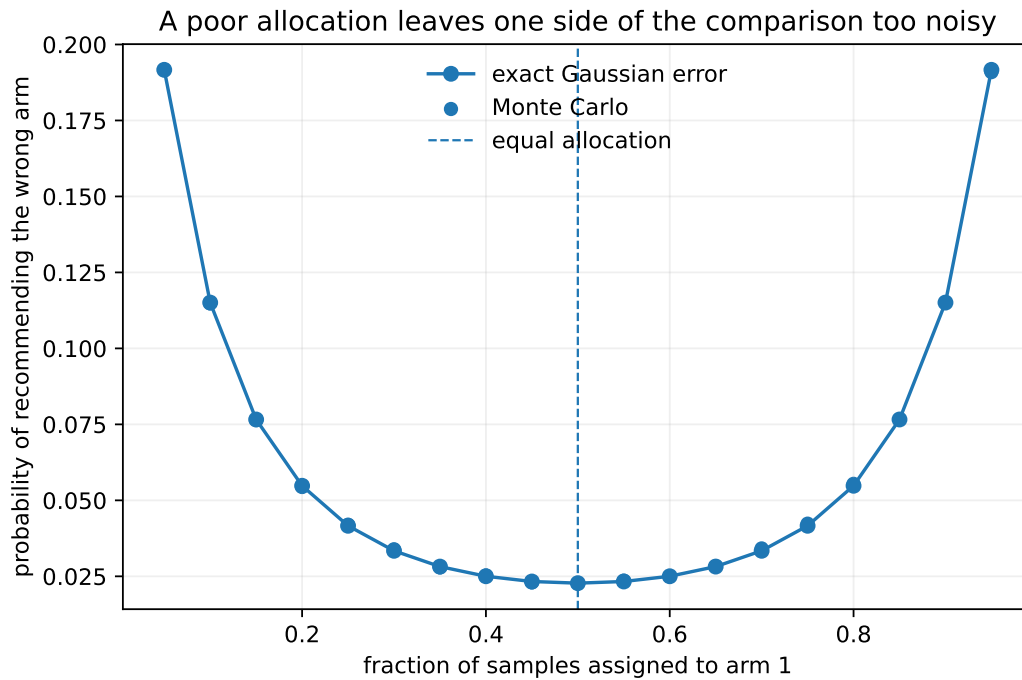


Figure 8.3: For $n = 400$, $(\mu_1, \mu_2) = (0.2, 0)$, and $\sigma = 1$, both exact calculation and Monte Carlo show that equal allocation minimizes the recommendation error.

Table 8.1: Selected Gaussian allocation results.

Fraction w on arm 1	exact error probability	information rate
0.10	0.1151	0.00180
0.30	0.0334	0.00420
0.50	0.0228	0.00500
0.70	0.0334	0.00420
0.90	0.1151	0.00180

```

1 n1 = max(1, int(round(total_budget * weight)))
2 n2 = total_budget - n1
3
4 mean1 = mu1 + sigma / np.sqrt(n1) * rng.standard_normal(replications)
5 mean2 = mu2 + sigma / np.sqrt(n2) * rng.standard_normal(replications)
6 empirical_error = np.mean(mean1 <= mean2)
7
8 standard_error = sigma * np.sqrt(1.0 / n1 + 1.0 / n2)
9 exact_error = norm.cdf(-(mu1 - mu2) / standard_error)

```


Chapter 9 A Glimpse of the Lai–Robbins Regret Lower Bound

The same argument also explains why logarithmic exploration is unavoidable in regret minimization.

Suppose arm a is suboptimal under ν :

$$\mu_a < \mu^*.$$

Construct an alternative ν' that changes only arm a , making it slightly better than μ^* . If the algorithm almost never samples arm a under ν , then the histories under ν and ν' remain too similar. But a good policy must behave differently:

- under ν , it should pull arm a rarely;
- under ν' , it should pull arm a almost all the time.

Choose an event such as

$$E_T = \left\{ N_a(T) < \frac{T}{2} \right\}.$$

For a sufficiently efficient policy,

$$\mathbb{P}_\nu(E_T) \rightarrow 1, \quad \mathbb{P}_{\nu'}(E_T) \rightarrow 0.$$

The transportation inequality gives

$$\mathbb{E}_\nu[N_a(T)] \text{KL}(\nu_a \parallel \nu'_a) \geq \text{kl}(\mathbb{P}_\nu(E_T), \mathbb{P}_{\nu'}(E_T)). \quad (9.1)$$

Under the consistency conditions used by Lai and Robbins, the right side grows like $\log T$. Moving ν'_a down toward the boundary where its mean equals μ^* yields

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}_\nu[N_a(T)]}{\log T} \geq \frac{1}{\text{KL}(\nu_a \parallel \nu^*)}. \quad (9.2)$$

Multiplying by the regret gap $\Delta_a = \mu^* - \mu_a$ and summing gives the classical shape

$$\liminf_{T \rightarrow \infty} \frac{R_T(\nu)}{\log T} \geq \sum_{a: \Delta_a > 0} \frac{\Delta_a}{\text{KL}(\nu_a \parallel \nu^*)}. \quad (9.3)$$

This section is only the proof skeleton. The next chapter will state the consistency assumptions carefully, construct the event with the correct polynomial probabilities, and derive the full Lai–Robbins lower bound without hiding the asymptotic steps.

Key idea

The logarithmic exploration rate is the price of ruling out a nearby world in which a currently suboptimal arm is actually optimal.

Chapter 10 What the Experiments Are Checking

The numerical work in this chapter serves as a proof audit rather than as evidence for the theorem.

10.1 Evidence paths

For $X_t \sim \text{Bern}(p)$, the cumulative evidence against q is

$$L_t = \sum_{s=1}^t \left[X_s \log \frac{p}{q} + (1 - X_s) \log \frac{1-p}{1-q} \right].$$

The experiment checks that paths are noisy while their mean slope is $\text{kl}(p, q)$.

10.2 Testing error

For each n , the script computes the exact sum of type-I and type-II errors of the equal-prior likelihood-ratio test between $\text{Bern}(0.4)^n$ and $\text{Bern}(0.6)^n$. It compares this with

$$\frac{1}{2} e^{-n \text{kl}(0.4, 0.6)}.$$

Table 10.1: Optimal testing error versus the Bretagnolle–Huber lower bound.

n	exact sum of errors	lower bound
10	0.5331	0.2222
20	0.3722	0.0988
40	0.2041	0.0195
80	0.0716	0.000761
160	0.0107	1.16×10^{-6}

The lower bound becomes loose in this example because it sacrifices constants for generality. It is still correct, and it preserves the central message that the error cannot decay independently of accumulated information.

10.3 Adaptive information accounting

The UCB experiment checks

$$\mathbb{E}_\nu[L_T] = \sum_a \mathbb{E}_\nu[N_a(T)] \text{KL}(\nu_a \| \nu'_a)$$

under an adaptive policy. The near overlap of the two curves is a direct numerical check that the likelihood-ratio implementation, pull counts, and KL formula agree.

10.4 Allocation and hard alternatives

The Gaussian experiment checks two linked predictions:

$$\text{hardest alternative for allocation } w \implies m_w = w\mu_1 + (1-w)\mu_2,$$

and

$$\text{best worst-case information rate} \implies w^* = \frac{1}{2}.$$

The exact finite-sample error confirms the same symmetry.

Think

A simulation can reveal a coding error or a missing factor of two. It cannot prove a universal lower bound. The proof ranges over all algorithms and all admissible alternatives; a simulation samples only finitely many histories from one implementation.

Chapter 11 Common Mistakes in Change-of-Measure Proofs

11.1 Choosing an alternative that does not change the answer

If $a^*(\lambda) = a^*(\nu)$, then the recommendation event need not have very different probabilities. The right side of (5.2) may be small, and no useful lower bound follows.

11.2 Choosing an alternative that is too far away

A dramatic alternative has a large KL cost per observation. Since the lower bound divides by this cost, the result becomes weak. The right alternative usually sits near the decision boundary.

11.3 Reversing the direction of KL

The expected log-likelihood ratio under ν is

$$\mathbb{E}_\nu \left[\log \frac{dP_\nu}{dP_{\nu'}} \right] = \text{KL}(P_\nu \| P_{\nu'}),$$

not the reverse divergence. The pull counts are also averaged under ν . Both directions must match.

11.4 Forgetting that counts are random

In an adaptive bandit, $N_a(T)$ depends on previous rewards. The correct identity uses

$$\mathbb{E}_\nu[N_a(T)] \text{KL}(\nu_a \| \nu'_a),$$

not a deterministic count inserted after the fact.

11.5 Using an event unavailable at the stopping time

The event E must belong to $\mathcal{F}_\tau$. A lower-bound proof cannot use future observations that the algorithm did not possess when it stopped.

11.6 Proving correctness only in the true environment

The argument needs the algorithm to be reliable in both ν and λ . This is why lower bounds assume uniform correctness over a model class. A procedure designed for one known instance could simply print the answer without observing anything.

11.7 Stopping after one convenient alternative

One alternative gives one constraint. A sharp lower bound minimizes over every answer-changing alternative and then optimizes the sampling allocation. The optimization is the theorem, not decorative notation.

Chapter 12 What to Carry Forward

The chapter can be compressed to five equations.

For two laws,

$$Q(E) = \mathbb{E}_P[\mathbf{1}_E e^{-L}], \quad L = \log \frac{dP}{dQ}.$$

Compressing an experiment to an event cannot increase information:

$$\text{KL}(P\|Q) \geq \text{kl}(P(E), Q(E)).$$

For a fixed-horizon adaptive bandit,

$$\text{KL}(P_\nu^T \| P_{\nu'}^T) = \sum_a \mathbb{E}_\nu[N_a(T)] \text{KL}(\nu_a \| \nu'_a).$$

For a stopped experiment and $E \in \mathcal{F}_\tau$,

$$\sum_a \mathbb{E}_\nu[N_a(\tau)] \text{KL}(\nu_a \| \nu'_a) \geq \text{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E)).$$

For fixed-confidence best-arm identification,

$$\frac{1}{T^*(\nu)} = \sup_{w \in \Delta_K} \inf_{\lambda \in \text{Alt}(\nu)} \sum_a w_a \text{KL}(\nu_a \| \lambda_a).$$

These equations express one idea in progressively richer settings:

To behave differently in two nearby worlds, a learner must observe enough of the places where those worlds differ.

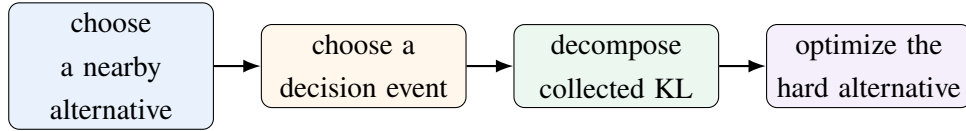


Figure 12.1: The reusable research pattern behind many bandit lower bounds.

Proof of the Bretagnolle–Huber Inequality

Let p and q be densities of P and Q with respect to a common measure. For any event E ,

$$\begin{aligned} P(E) + Q(E^c) &= \int_E p + \int_{E^c} q \\ &\geq \int \min\{p, q\}. \end{aligned}$$

Define the affinity

$$\rho(P, Q) = \int \sqrt{pq}.$$

By Cauchy–Schwarz,

$$\begin{aligned} \rho(P, Q)^2 &= \left(\int \sqrt{\min\{p, q\}} \sqrt{\max\{p, q\}} \right)^2 \\ &\leq \left(\int \min\{p, q\} \right) \left(\int \max\{p, q\} \right). \end{aligned}$$

Since

$$\int \max\{p, q\} = 2 - \int \min\{p, q\} \leq 2,$$

we obtain

$$\int \min\{p, q\} \geq \frac{\rho(P, Q)^2}{2}. \quad (1)$$

Next relate affinity to KL. Under P ,

$$\begin{aligned} \rho(P, Q) &= \int p \sqrt{\frac{q}{p}} \\ &= \mathbb{E}_P \left[\sqrt{\frac{q}{p}} \right]. \end{aligned}$$

Because $-\log$ is convex, Jensen's inequality gives

$$\begin{aligned} -\log \rho(P, Q) &\leq \mathbb{E}_P \left[-\log \sqrt{\frac{q}{p}} \right] \\ &= \frac{1}{2} \mathbb{E}_P \left[\log \frac{p}{q} \right] \\ &= \frac{1}{2} \text{KL}(P \| Q). \end{aligned}$$

Therefore

$$\rho(P, Q) \geq \exp \left\{ -\frac{1}{2} \text{KL}(P \| Q) \right\}.$$

Squaring and substituting into (1),

$$\int \min\{p, q\} \geq \frac{1}{2} e^{-\text{KL}(P \| Q)}.$$

Finally,

$$P(E) + Q(E^c) \geq \int \min\{p, q\} \geq \frac{1}{2} e^{-\text{KL}(P \| Q)}.$$

A Direct Jensen Proof of Event Compression

The change-of-measure identity gives

$$Q(E) = \mathbb{E}_P[\mathbf{1}_E e^{-L}].$$

Condition on E :

$$Q(E) = P(E) \mathbb{E}_P[e^{-L} \mid E].$$

Jensen's inequality for the convex exponential function yields

$$\mathbb{E}_P[e^{-L} \mid E] \geq e^{-\mathbb{E}_P[L \mid E]}.$$

Hence

$$Q(E) \geq P(E) e^{-\mathbb{E}_P[L \mid E]},$$

which rearranges to

$$\mathbb{E}_P[L \mid E] \geq \log \frac{P(E)}{Q(E)}.$$

The same argument on E^c gives

$$\mathbb{E}_P[L \mid E^c] \geq \log \frac{P(E^c)}{Q(E^c)}.$$

Therefore

$$\begin{aligned} \mathbb{E}_P[L] &= P(E) \mathbb{E}_P[L \mid E] + P(E^c) \mathbb{E}_P[L \mid E^c] \\ &\geq P(E) \log \frac{P(E)}{Q(E)} + P(E^c) \log \frac{P(E^c)}{Q(E^c)} \\ &= \text{kl}(P(E), Q(E)). \end{aligned}$$

Since $\mathbb{E}_P[L] = \text{KL}(P \parallel Q)$, this proves (3.2).

Formula Sheet

Table 1: Core formulas for change-of-measure lower bounds.

Object	Formula
Log-likelihood ratio	$L = \log(dP/dQ)$
Change of measure	$Q(E) = \mathbb{E}_P[\mathbf{1}_E e^{-L}]$
Binary data processing	$\text{KL}(P\ Q) \geq \text{kl}(P(E), Q(E))$
Bretagnolle–Huber	$P(E) + Q(E^c) \geq \frac{1}{2} e^{-\text{KL}(P\ Q)}$
Bandit history LLR	$L_T = \sum_{t=1}^T \log[p_{A_t}(X_t)/p'_{A_t}(X_t)]$
Divergence decomposition	$\text{KL}(P_\nu^T\ P_{\nu'}^T) = \sum_a \mathbb{E}_\nu[N_a(T)] \text{KL}(\nu_a\ \nu'_a)$
Stopped transportation	$\sum_a \mathbb{E}_\nu[N_a(\tau)] \text{KL}(\nu_a\ \nu'_a) \geq \text{kl}(\mathbb{P}_\nu(E), \mathbb{P}_{\nu'}(E))$
Fixed-confidence demand	$\text{kl}(1 - \delta, \delta) \sim \log(1/\delta)$
BAI information value	$T^*(\nu)^{-1} = \sup_w \inf_{\lambda \in \text{Alt}(\nu)} \sum_a w_a \text{KL}(\nu_a\ \lambda_a)$
Two-Gaussian rate	$T^*(\nu)^{-1} = \Delta^2/(8\sigma^2)$

Notation Table

Table 2: Notation used in this chapter.

Symbol	Meaning
P, Q	two probability laws being compared
L	log-likelihood ratio $\log(dP/dQ)$
E	a measurable decision event
$\text{kl}(x, y)$	Bernoulli or binary relative entropy
ν, ν'	two bandit environments
π_t	policy kernel at round t
H_t	observed history through round t
A_t, X_t	selected arm and observed reward at round t
$N_a(T)$	number of pulls of arm a by time T
τ	stopping time
$\mathcal{F}_\tau$	information available when the algorithm stops
$\text{Alt}(\nu)$	environments whose correct answer differs from that under ν
w	vector of expected sampling proportions
$T^*(\nu)$	characteristic information time for best-arm identification
Δ	gap between the two arm means

Implementation Notes

The supplied Python script regenerates all figures and the CSV table. Several details are worth checking when adapting it:

1. Keep the direction of the likelihood ratio consistent: the simulation draws from the true environment and computes $\log(p_\nu/p_{\nu'})$.
2. When an arm is unchanged between environments, its likelihood-ratio increment is exactly zero.
3. Initialize every arm before evaluating a UCB index.
4. In Gaussian allocation experiments, drawing the sample mean directly is exactly equivalent to simulating every individual observation and is much faster.
5. Compare Monte Carlo output with an analytic identity whenever one is available. Here the exact Gaussian error and the divergence decomposition provide two independent audits.

Complete Reproducible Code

```
1  """Reproducible experiments for
2  Change of Measure Arguments in Bandit Lower Bounds.
3
4  The script creates seven figures and one CSV file in the same directory.
5  All random experiments use a fixed NumPy seed.
6  """
7
8  from __future__ import annotations
9
10 import csv
11 import math
12 from pathlib import Path
13 from typing import Iterable
14
15 import matplotlib.pyplot as plt
16 import numpy as np
17 from scipy.stats import binom, norm
18
19 OUTPUT_DIR = Path(__file__).resolve().parent
20 SEED = 20260620
21
22
23 def bernoulli_kl(p: np.ndarray | float, q: np.ndarray | float) -> np.ndarray | float:
24     """KL(Bernoulli(p) || Bernoulli(q)), with stable clipping."""
25     eps = np.finfo(float).eps
26     p_arr = np.clip(np.asarray(p, dtype=float), eps, 1.0 - eps)
27     q_arr = np.clip(np.asarray(q, dtype=float), eps, 1.0 - eps)
28     value = p_arr * np.log(p_arr / q_arr) + (1.0 - p_arr) * np.log(
29         (1.0 - p_arr) / (1.0 - q_arr)
30     )
31     if np.ndim(value) == 0:
32         return float(value)
33     return value
34
35
36 def bernoulli_llr(reward: np.ndarray, p: np.ndarray, q: np.ndarray) -> np.ndarray:
37     """log p(X)/q(X) for Bernoulli rewards and possibly arm-dependent means."""
38     return reward * np.log(p / q) + (1.0 - reward) * np.log((1.0 - p) / (1.0 - q))
39
40
41 def save_figure(fig: plt.Figure, stem: str) -> None:
42     fig.savefig(OUTPUT_DIR / f"{stem}.pdf", bbox_inches="tight")
43     fig.savefig(OUTPUT_DIR / f"{stem}.png", dpi=220, bbox_inches="tight")
44     plt.close(fig)
45
46
47 def experiment_evidence_paths(rng: np.random.Generator, rows: list[dict[str, object]]) -> None:
48     p, q = 0.55, 0.45
49     horizon = 400
50     paths = 8
51     rewards = rng.binomial(1, p, size=(paths, horizon)).astype(float)
52     increments = bernoulli_llr(rewards, np.full_like(rewards, p), np.full_like(rewards, q))
53     cumulative = np.cumsum(increments, axis=1)
54     expected = np.arange(1, horizon + 1) * bernoulli_kl(p, q)
55
56     fig = plt.figure(figsize=(7.2, 4.5))
57     ax = fig.add_subplot(111)
```

```

58     t = np.arange(1, horizon + 1)
59     for path in cumulative:
60         ax.plot(t, path, linewidth=1.0, alpha=0.72)
61     ax.plot(t, expected, linewidth=2.4, linestyle="--", label=r"$t\backslash, \mathrm{kl}(p,q)$")
62     ax.axhline(0.0, linewidth=0.8)
63     ax.set_xlabel("number of observations")
64     ax.set_ylabel("cumulative log-likelihood ratio")
65     ax.set_title("Evidence is noisy, but its mean drift is KL")
66     ax.legend(frameon=False)
67     ax.grid(alpha=0.2)
68     save_figure(fig, "evidence_paths")
69
70     rows.append(
71         {
72             "experiment": "evidence_paths",
73             "x": horizon,
74             "setting": f"Bernoulli({p}) vs Bernoulli({q})",
75             "metric": "mean_final_llr",
76             "value": float(cumulative[:, -1].mean()),
77         }
78     )
79     rows.append(
80         {
81             "experiment": "evidence_paths",
82             "x": horizon,
83             "setting": f"Bernoulli({p}) vs Bernoulli({q})",
84             "metric": "theoretical_mean_final_llr",
85             "value": float(expected[-1]),
86         }
87     )
88
89
90 def exact_lrt_error_sum(n: int, p: float, q: float) -> float:
91     """Sum of the two errors for the equal-prior likelihood-ratio test."""
92     successes = np.arange(n + 1)
93     llr = successes * math.log(p / q) + (n - successes) * math.log((1.0 - p) / (1.0 - q))
94     choose_p = llr >= 0.0
95     error_under_p = binom.pmf(successes, n, p)[~choose_p].sum()
96     error_under_q = binom.pmf(successes, n, q)[choose_p].sum()
97     return float(error_under_p + error_under_q)
98
99
100 def experiment_testing_bound(rows: list[dict[str, object]]) -> None:
101     p, q = 0.40, 0.60
102     sample_sizes = np.array([5, 10, 20, 40, 80, 160, 240])
103     exact_errors = np.array([exact_lrt_error_sum(int(n), p, q) for n in sample_sizes])
104     bh_bounds = 0.5 * np.exp(-sample_sizes * bernoulli_kl(p, q))
105
106     fig = plt.figure(figsize=(7.2, 4.5))
107     ax = fig.add_subplot(111)
108     ax.semilogy(sample_sizes, exact_errors, marker="o", label="optimal likelihood-ratio test")
109     ax.semilogy(sample_sizes, bh_bounds, marker="s", linestyle="--", label="Bretagnolle-Huber lower bound")
110     ax.set_xlabel("sample size n")
111     ax.set_ylabel("sum of the two error probabilities")
112     ax.set_title("Nearby models cannot be separated faster than their information allows")
113     ax.legend(frameon=False)
114     ax.grid(alpha=0.2)
115     save_figure(fig, "testing_error_bound")
116
117     for n, exact, bound in zip(sample_sizes, exact_errors, bh_bounds):
118         rows.extend(

```

```

119     [
120         {
121             "experiment": "testing_bound",
122             "x": int(n),
123             "setting": f"Bernoulli({p}) vs Bernoulli({q})",
124             "metric": "exact_lrt_error_sum",
125             "value": float(exact),
126         },
127         {
128             "experiment": "testing_bound",
129             "x": int(n),
130             "setting": f"Bernoulli({p}) vs Bernoulli({q})",
131             "metric": "bretagnolle_huber_bound",
132             "value": float(bound),
133         },
134     ]
135 )
136
137
138 def simulate_ucb_information(
139     rng: np.random.Generator,
140     horizon: int,
141     replications: int,
142     true_means: np.ndarray,
143     alternative_means: np.ndarray,
144 ) -> tuple[np.ndarray, np.ndarray]:
145     """Run UCB under the true environment and return counts and total LLR."""
146     arms = len(true_means)
147     counts = np.zeros((replications, arms), dtype=np.int64)
148     sums = np.zeros((replications, arms), dtype=float)
149     total_llr = np.zeros(replications, dtype=float)
150     indices = np.arange(replications)
151
152     # Pull each arm once.
153     for arm in range(arms):
154         rewards = rng.binomial(1, true_means[arm], size=replications).astype(float)
155         counts[:, arm] += 1
156         sums[:, arm] += rewards
157         total_llr += bernoulli_llr(
158             rewards,
159             np.full(replications, true_means[arm]),
160             np.full(replications, alternative_means[arm]),
161         )
162
163     for t in range(arms, horizon):
164         means = sums / counts
165         bonus = np.sqrt(2.0 * np.log(t + 1.0) / counts)
166         actions = np.argmax(means + bonus, axis=1)
167         arm_means = true_means[actions]
168         alt_means = alternative_means[actions]
169         rewards = rng.binomial(1, arm_means).astype(float)
170         counts[indices, actions] += 1
171         sums[indices, actions] += rewards
172         total_llr += bernoulli_llr(rewards, arm_means, alt_means)
173
174     return counts, total_llr
175
176
177 def experiment_adaptive_information(rng: np.random.Generator, rows: list[dict[str, object]]) -> None:
178     true_means = np.array([0.62, 0.50])
179     alternative_means = np.array([0.62, 0.68])

```

```

180 horizons = np.array([50, 100, 200, 400, 800, 1200])
181 replications = 12000
182 empirical_llr: list[float] = []
183 accounted_information: list[float] = []
184 avg_arm2_counts: list[float] = []
185 arm_kls = np.asarray(bernoulli_kl(true_means, alternative_means))
186
187 for horizon in horizons:
188     counts, total_llr = simulate_ucb_information(
189         rng,
190         int(horizon),
191         replications,
192         true_means,
193         alternative_means,
194     )
195     average_counts = counts.mean(axis=0)
196     lhs = float(total_llr.mean())
197     rhs = float(np.dot(average_counts, arm_kls))
198     empirical_llr.append(lhs)
199     accounted_information.append(rhs)
200     avg_arm2_counts.append(float(average_counts[1]))
201     rows.extend(
202         [
203             {
204                 "experiment": "adaptive_information",
205                 "x": int(horizon),
206                 "setting": "UCB: true=(0.62,0.50), alternative=(0.62,0.68)",
207                 "metric": "empirical_mean_log_likelihood_ratio",
208                 "value": lhs,
209             },
210             {
211                 "experiment": "adaptive_information",
212                 "x": int(horizon),
213                 "setting": "UCB: true=(0.62,0.50), alternative=(0.62,0.68)",
214                 "metric": "sum_expected_pulls_times_arm_kl",
215                 "value": rhs,
216             },
217             {
218                 "experiment": "adaptive_information",
219                 "x": int(horizon),
220                 "setting": "UCB: true=(0.62,0.50), alternative=(0.62,0.68)",
221                 "metric": "average_pulls_arm_2",
222                 "value": float(average_counts[1]),
223             },
224         ]
225     )
226
227 fig = plt.figure(figsize=(7.2, 4.5))
228 ax = fig.add_subplot(111)
229 ax.plot(horizons, empirical_llr, marker="o", label="empirical mean log-likelihood ratio")
230 ax.plot(horizons, accounted_information, marker="s", linestyle="--", label=r"$\sum_a \mathbb{E}[N_a]D(\nu_a, \nu'_a)$")
231 ax.set_xlabel("horizon")
232 ax.set_ylabel("information")
233 ax.set_title("Adaptive sampling changes the counts, not the information ledger")
234 ax.legend(frameon=False)
235 ax.grid(alpha=0.2)
236 save_figure(fig, "adaptive_information_identity")
237
238 fig = plt.figure(figsize=(7.2, 4.5))
239 ax = fig.add_subplot(111)

```

```

240 ax.plot(horizons, avg_arm2_counts, marker="o")
241 ax.set_xlabel("horizon")
242 ax.set_ylabel("average pulls of the distinguishing arm")
243 ax.set_title("UCB gathers evidence by revisiting the arm changed in the alternative")
244 ax.grid(alpha=0.2)
245 save_figure(fig, "distinguishing_arm_counts")
246
247
248 def experiment_gaussian_allocation(rng: np.random.Generator, rows: list[dict[str, object]]) -> None:
249     mu1, mu2, sigma = 0.20, 0.00, 1.0
250     total_budget = 400
251     replications = 150000
252     weights = np.arange(0.05, 1.00, 0.05)
253     empirical_errors: list[float] = []
254     exact_errors: list[float] = []
255     information_rates: list[float] = []
256
257     for weight in weights:
258         n1 = max(1, int(round(total_budget * weight)))
259         n2 = total_budget - n1
260         mean1 = mu1 + sigma / math.sqrt(n1) * rng.standard_normal(replications)
261         mean2 = mu2 + sigma / math.sqrt(n2) * rng.standard_normal(replications)
262         empirical = float(np.mean(mean1 <= mean2))
263         standard_error = sigma * math.sqrt(1.0 / n1 + 1.0 / n2)
264         exact = float(norm.cdf(-(mu1 - mu2) / standard_error))
265         effective_weight = n1 / total_budget
266         information_rate = effective_weight * (1.0 - effective_weight) * (mu1 - mu2) ** 2 / (
267             2.0 * sigma**2
268         )
269         empirical_errors.append(empirical)
270         exact_errors.append(exact)
271         information_rates.append(information_rate)
272         rows.extend(
273             [
274                 {
275                     "experiment": "gaussian_allocation",
276                     "x": effective_weight,
277                     "setting": f"mu=({mu1},{mu2}), sigma={sigma}, budget={total_budget}",
278                     "metric": "empirical_error_probability",
279                     "value": empirical,
280                 },
281                 {
282                     "experiment": "gaussian_allocation",
283                     "x": effective_weight,
284                     "setting": f"mu=({mu1},{mu2}), sigma={sigma}, budget={total_budget}",
285                     "metric": "exact_error_probability",
286                     "value": exact,
287                 },
288                 {
289                     "experiment": "gaussian_allocation",
290                     "x": effective_weight,
291                     "setting": f"mu=({mu1},{mu2}), sigma={sigma}, budget={total_budget}",
292                     "metric": "hardest_alternative_information_rate",
293                     "value": information_rate,
294                 },
295             ]
296         )
297
298 fig = plt.figure(figsize=(7.2, 4.5))
299 ax = fig.add_subplot(111)
300 ax.plot(weights, exact_errors, marker="o", label="exact Gaussian error")

```

```

301 ax.scatter(weights, empirical_errors, s=28, label="Monte Carlo")
302 ax.axvline(0.5, linestyle="--", linewidth=1.0, label="equal allocation")
303 ax.set_xlabel("fraction of samples assigned to arm 1")
304 ax.set_ylabel("probability of recommending the wrong arm")
305 ax.set_title("A poor allocation leaves one side of the comparison too noisy")
306 ax.legend(frameon=False)
307 ax.grid(alpha=0.2)
308 save_figure(fig, "gaussian_allocation_error")
309
310 fig = plt.figure(figsize=(7.2, 4.5))
311 ax = fig.add_subplot(111)
312 ax.plot(weights, information_rates, marker="o")
313 ax.axvline(0.5, linestyle="--", linewidth=1.0)
314 ax.set_xlabel("fraction of samples assigned to arm 1")
315 ax.set_ylabel("information rate against the hardest alternative")
316 ax.set_title("The lower-bound game selects equal allocation in the symmetric Gaussian case")
317 ax.grid(alpha=0.2)
318 save_figure(fig, "gaussian_information_rate")
319
320
321 def experiment_alternative_geometry(rows: list[dict[str, object]]) -> None:
322     mu1, mu2, sigma = 0.20, 0.00, 1.0
323     means = np.linspace(-0.08, 0.28, 361)
324     weights: Iterable[float] = (0.20, 0.50, 0.80)
325
326     fig = plt.figure(figsize=(7.2, 4.5))
327     ax = fig.add_subplot(111)
328     for weight in weights:
329         objective = (
330             weight * (mu1 - means) ** 2 + (1.0 - weight) * (mu2 - means) ** 2
331         ) / (2.0 * sigma**2)
332         minimizer = weight * mu1 + (1.0 - weight) * mu2
333         minimum = weight * (1.0 - weight) * (mu1 - mu2) ** 2 / (2.0 * sigma**2)
334         ax.plot(means, objective, label=f"allocation w={weight:.1f}")
335         ax.scatter([minimizer], [minimum], s=35)
336         rows.extend(
337             [
338                 {
339                     "experiment": "alternative_geometry",
340                     "x": weight,
341                     "setting": f"Gaussian mu=({mu1},{mu2}), sigma={sigma}",
342                     "metric": "hardest_common_mean",
343                     "value": minimizer,
344                 },
345                 {
346                     "experiment": "alternative_geometry",
347                     "x": weight,
348                     "setting": f"Gaussian mu=({mu1},{mu2}), sigma={sigma}",
349                     "metric": "minimum_information_rate",
350                     "value": minimum,
351                 },
352             ]
353         )
354     ax.set_xlabel("common mean m in the boundary alternative")
355     ax.set_ylabel(r"$wD(\mu_1, m) + (1-w)D(\mu_2, m)$")
356     ax.set_title("The hardest alternative moves toward the least-observed arm")
357     ax.legend(frameon=False)
358     ax.grid(alpha=0.2)
359     save_figure(fig, "gaussian_alternative_geometry")
360
361

```



```

362 def write_results(rows: list[dict[str, object]]) -> None:
363     output = OUTPUT_DIR / "change_of_measure_results.csv"
364     with output.open("w", newline="", encoding="utf-8") as handle:
365         writer = csv.DictWriter(handle, fieldnames=["experiment", "x", "setting", "metric", "value"])
366         writer.writeheader()
367         writer.writerows(rows)
368
369
370 def main() -> None:
371     rng = np.random.default_rng(SEED)
372     rows: list[dict[str, object]] = []
373     experiment_evidence_paths(rng, rows)
374     experiment_testing_bound(rows)
375     experiment_adaptive_information(rng, rows)
376     experiment_gaussian_allocation(rng, rows)
377     experiment_alternative_geometry(rows)
378     write_results(rows)
379     print(f"Wrote figures and results to {OUTPUT_DIR}")
380
381
382 if __name__ == "__main__":
383     main()

```

Further Reading

The divergence-decomposition presentation in Lattimore and Szepesvari's *Bandit Algorithms* is an especially clear route from information theory to bandit lower bounds [7]. Kaufmann, Cappe, and Garivier give a stopped change-of-distribution inequality tailored to best-arm identification and use it to derive instance-dependent complexity bounds [5]. Garivier and Kaufmann turn the resulting allocation game into the Track-and-Stop algorithm and a tight asymptotic characterization for one-parameter bandits [3].

The regret direction begins with Lai and Robbins [6] and its multiparameter extension by Burnetas and Katehakis [1]. For the broader testing tradition behind these arguments, see Chernoff's sequential design work [2], Wald's sequential analysis [11], and modern lower-bound treatments by Tsybakov [10] and Le Cam and Yang [8]. Recent work by Yang, Tan, and Jin uses information-theoretic lower bounds to study best-arm identification with minimal cumulative regret [12], while Jin and collaborators study how near-optimal information allocation can be implemented under batching constraints [4].

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